
# Designing Cyber-Safe Connected Vehicle Architecture

Whitepaper

# Abstract:

Connected cars, much like smartphones, have ushered in the era of disruption in the auto industry. It has made us reimagine the vehicles we want to drive. This change will be sooner than we imagine. Connected cars bring huge possibilities, but they also open a significant number of avenues through which a vehicle can be hacked with the possibility of stealing personal information. Chipset makers, technology providers, vehicle manufacturers (OEM), and Tier-1 companies are continuously trying to strengthen the vehicle architecture without compromising the functionalities. This whitepaper gives an overview of various connectivity interfaces of connected vehicles, their possible vulnerabilities, and changing vehicle architecture.

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# Introduction

Till early 2000, cars were seen as a luxury for ordinary people in developing countries. Gradually, four-wheelers are now the essential means of commuting and an integral part of our daily life. About 70 million cars are sold globally annually. The two major markets are China, which accounts for approximately 21 million units, and the USA, which buys approximately 15 million units. The evolution of technologies including connectivity, computing, cloud, sensor fusion, and data processing has opened many vistas for enhancing passenger comfort and safety. The car we drive today is sophisticated and complex—a place where multiple technologies converge. Much like smartphones, cars are a hub for connectivity. Cars have already been called 'Cell on wheels', alluding to the different wireless technologies, namely cellular – LTE/5G, Wi-Fi, Bluetooth, NFC, Ultra-Wide Band, among others. Wired connectivity based on technologies such as CAN, Ethernet, MOST, USB, etc., continues to be an integral part of the car. The flip side is that all these different modes of connectivity offer several vistas for security vulnerabilities. The range of threats includes theft, unauthorized access of the vehicle, especially control of critical vehicle functions such as brakes, engine, steering, immobilizer etc.

# Connectivity interfaces and uses

The various connectivity interfaces provide different ways to access in-vehicle devices such as Electronic Control Units (ECUs) and applications. As there are multiple interfaces there, the key challenge is that there is no single point that can be made secure and preserve the integrity of the vehicle's electronically controlled functions.

Connectivity interfaces in a vehicle may be categorized into internal connectivity, short-range connectivity, and long-range or external connectivity. Figure 2 shows the internal linkages among the different connectivity interfaces.

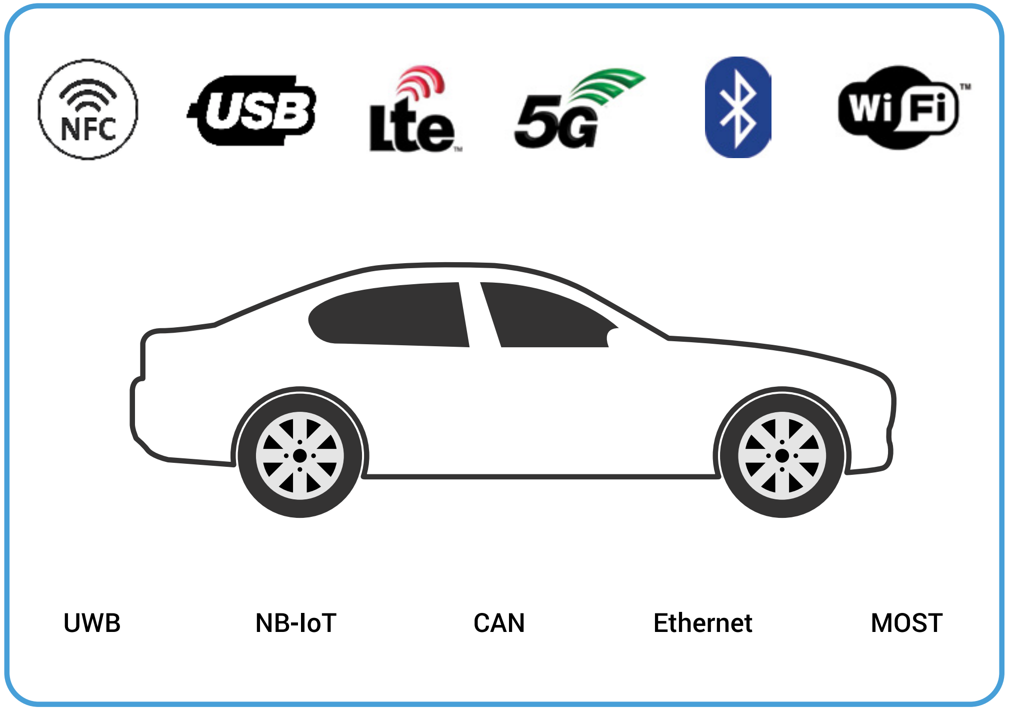



Figure 1: Vehicle Connectivity Interfaces

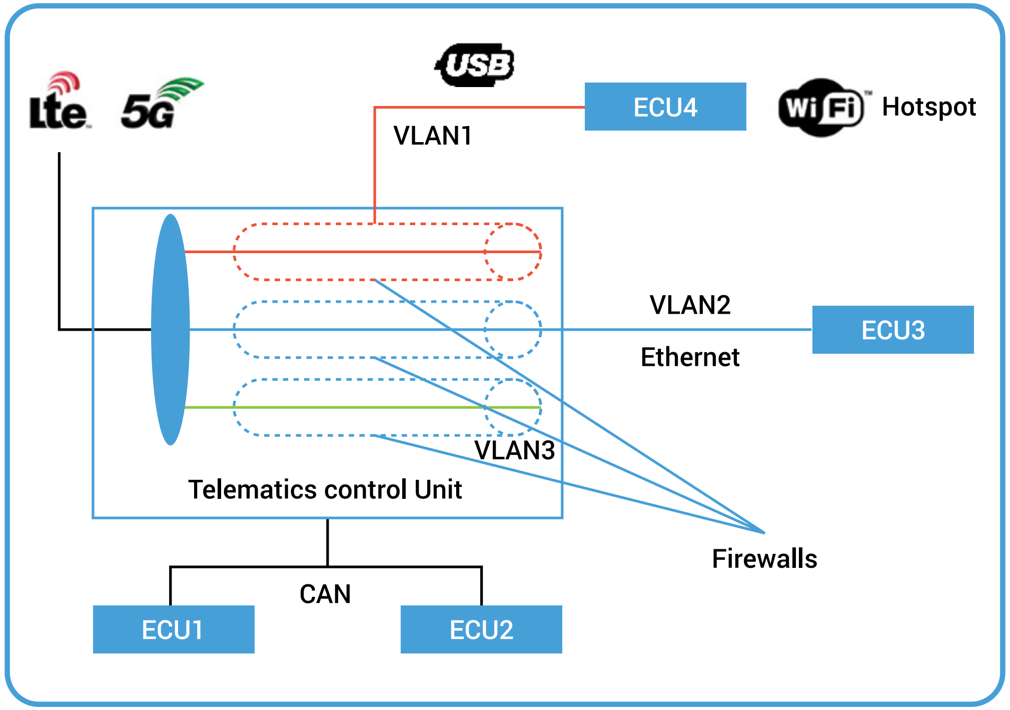
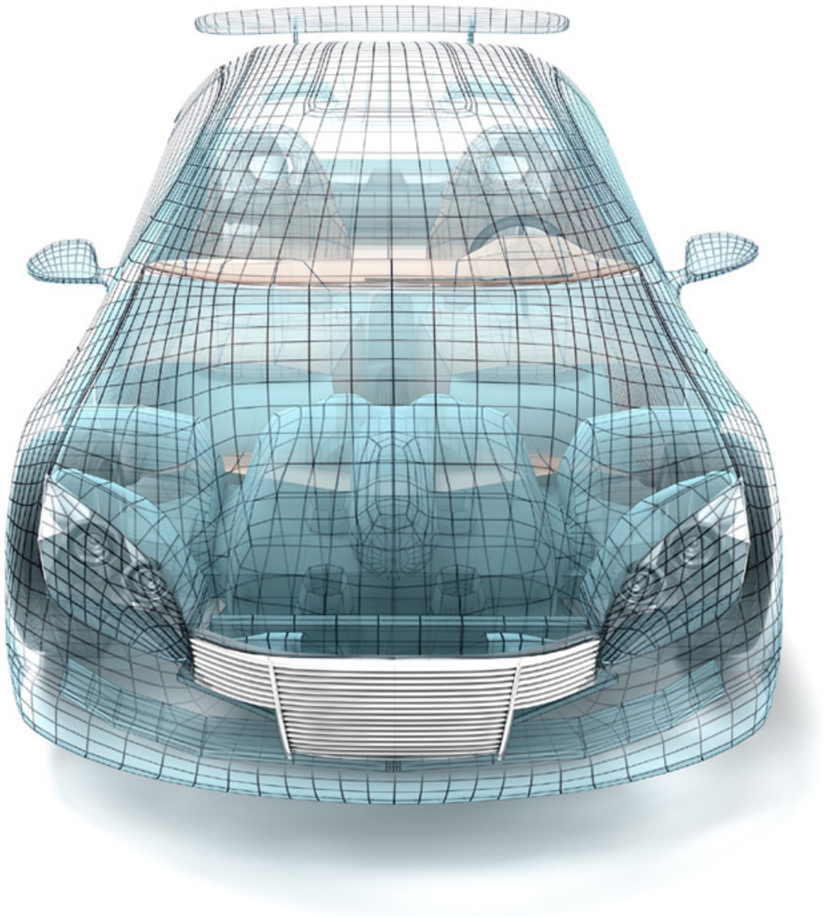



Figure 2: Telematics Control Unit with data endpoints

In the above figure 2, the Telematics Control Unit (TCU) connects externally over 4G/LTE or 5G networks and provides internal data endpoints as Virtual LANs (VLANs). Each of the endpoints is protected with firewalls. The TCU connects to other ECUs over USB and provides virtual ethernet endpoints for data sharing between ECU4 (as shown in the illustration). The same ECU4 may provide a Wi-Fi hotspot supporting wireless data endpoints. Additionally, ECU4 may share data over Ethernet with ECU3 and can work as a gateway. The TCU is also connected over the CAN bus and communicates with other ECUs in the vehicle for different services such as emergency calls, stolen vehicle tracking, remote start, door lock/unlock, remote diagnostics, and many other features. In this configuration, the TCU is the nerve center controlling many critical functions in the connected vehicle. Over time, a vast number of applications from different ECUs will use the data links provided by the TCU. Thus, the TCU is the backbone, playing the roles of a data gateway and connectivity hub. This opens the possibility of the connected car being vulnerable to threats posed by the various connectivity interfaces. TCUs are computationally intensive and have inbuilt security and safety solutions. They are becoming increasingly difficult to manage.

# Internal Connectivity

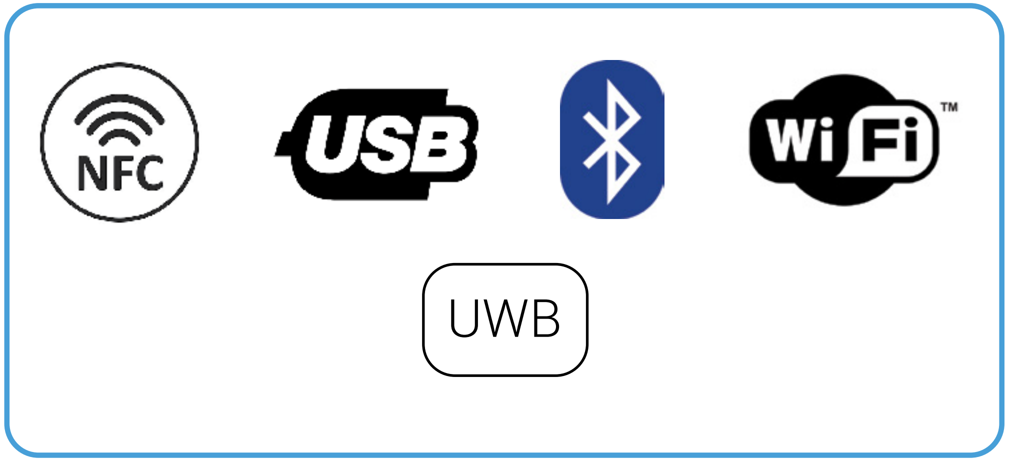


Wired interfaces such as CAN, Ethernet, and MOST are mainly used for internal vehicle connectivity. In current generation vehicles, there are more than 150 Electronics Control Units (ECUs) or Embedded Computing Units serving different functional needs of the car. These ECUs communicate with each other over CAN, Ethernet, MOST, or LIN. For the most part, ECUs communicate vehicle specific data, status parameters from various sensors and actuators, and control commands for actuators. Communication over these modes is typically in raw format without the use of encryption. Some of the vehicle manufacturers are gradually moving towards encrypted internal communication (Secure Onboard Communication) supported by standards defined and published by AUTOSAR.org.

**Vulnerabilities:**

Raw communication exposes the data to vulnerabilities. This exposes the system to the possibility of unauthorized attempts to modify control commands for actuators or generating catastrophic errors on the bus, which results in stopping the communication altogether. In the realm of wired connectivity, vulnerability is limited to those having physical access to the vehicle, including the wiring harness, and possessing a detailed knowledge of vehicle communication message formats.

# Short Range Connectivity




Interfaces such as Wi-Fi, Bluetooth, NFC, USB, UWB, and V2X are used for short-range to very short-range connectivity. Some of the interfaces, such as NFC and UWB, are used for specific purposes only, and they do not pose significant vulnerabilities due to their limited usage. One of the common uses of NFC is to enable easy and secure pairing of mobile devices with In-Vehicle Infotainment (IVI) systems. Connectivity using UWB (Ultra-Wide Band) technology is suitable to provide precise location detection, which can support multiple use cases. The Angle-of-Arrival of the user enables a particular ECU inside the vehicle to unlock the vehicle door based on the proximity of the user. A similar action can be taken to turn on the lights, enabling other use cases such as exchanges of the Car Connectivity Consortium's (CCC) Digital Key.

USB provides a one-to-one communication interface between the smartphone and the IVI System or between two ECUs. USB communication normally emulates ethernet interfaces (NCM, ECM, EEM or similar) and provides communication channels which can be used for sharing data, supporting projection mode such as CarPlay or Android Auto, as the case may be.

Bluetooth (BT) is a popular connectivity mechanism used by many handheld devices. In a vehicle, BT, or BT Low Energy (BLE) based connectivity is used for many standard and complex use cases. Bluetooth provides handsfree connectivity, enabling the user to use their phone while driving without minimal distraction. The applications supported include voice calling, text to speech or speech to text, sharing address books, and various other applications such as music streaming, social networking etc. In addition to supporting handsfree communication, BT and BLE are becoming critical connectivity interfaces for CCC Digital Key Exchange. In such use cases, the user's smart device serves as a Phone as a Key (Palak) for the vehicle and involves authentication, authorization, and many other secure interchanges. The Digital Key is highly secure and opens the paradigm of sharing your key with an unlimited number of users with different levels of restriction, allowing partial access to the vehicle. A Bluetooth-based Personal Area Network is also used to share data from one ECU to a phone or other device.

Wi-Fi is another common interface available inside the vehicle. An ECU (normally an IVI unit) may use Wi-Fi to provide the functionality of a Wi-Fi hotspot. In the scenario, the hotspot enables connection to a VLAN shared by the TCU. This connectivity interface can be used by user devices such as smartphones, tablets, laptops, etc. In addition to these use cases, other ECUs may use the data connection for some specific applications.

**Vulnerabilities:**

NFC or UWB are not yet used for wide use cases, and hence communication is very restricted. With restricted communication, they do not pose a significant vulnerability with regards to the whole vehicle.

USB is used for one-to-one connectivity where two ECUs can share the data channel using virtual ethernet interfaces. Any IP network poses a significant risk if it is not protected adequately through firewalls or other security mechanisms.

Bluetooth is a wireless technology, and there are many ways hackers can intrude on Bluetooth connections between the phone and ECUs. With the introduction of private-public key infrastructure, end-to-end encrypted communication ensures protection from such attacks, but being a wireless network, hackers can still disrupt the communication by using external hardware equipment.

Wi-Fi provides easy access to the in-car network as this is meant to be used by any occupant of the car. This calls for appropriate separation of the Wi-Fi based VLAN from the other VLANs such that critical data paths are not compromised.

# External Connectivity

The TCU provides external connectivity via access to cellular networks such as 4G/LTE or 5G. At this point, the number of TCU devices supporting 5G connectivity is still low. The lag in adoption is because OEMs are yet to discover the potential use cases that will motivate the migration from 4G to 5G. In the case of 4G, there are different Access Point Names (APNs) which provide separate data endpoints as shown in figure 2. OEMs, along with the network service providers, have created private APNs which enable isolated (secure) network connections between the TCU and the backend. This private network is not visible to other devices such as mobiles that use the data service from the same network provider. These private APNs are used for sending secure data to specific server ports and blocking access from any other IP or ports. This type of connection is used for safety applications such as emergency calls, stolen vehicle tracking, remote diagnostics, remote control of the car, etc. There may be other APNs which provide access to the public internet for applications running on various ECUs. Separate data networks create separate VLANs, which are protected from each other by firewalls. In the case of 5G, there are separate IP endpoints that provide separate VLANs.

**Vulnerabilities:**

The TCU communicates with the network service provider to establish multiple IP endpoints. Each of the endpoints is separated from one another using firewalls. But if the TCU is managing all the firewalls and data protection on its own, as shown in Figure 2, there is a possibility that hackers penetrate the networks through the TCU and create holes in the firewall. This is because most of the TCUs run embedded Linux and the kernels are not updated with the latest security patches and are not tested for all possible vulnerabilities.

# Change in Connected Vehicle Architecture

As discussed above, the TCU is the backbone of connectivity, and it is becoming increasingly complex to address various cyberthreats and vulnerabilities. Also, the number of ECUs and applications using the data network is increasing exponentially, which calls for robust network security measures to be implemented while supporting enhanced performance. Also, complex applications necessitate that the entire vehicle software stay updated through a remotely enabled mechanism for most of the ECUs. In a modern connected vehicle, even the body control system runs distributed applications directly from the cloud based on stored user preference.

To address these challenges, vehicle OEMs are proposing new architectures as below in Figure 3.

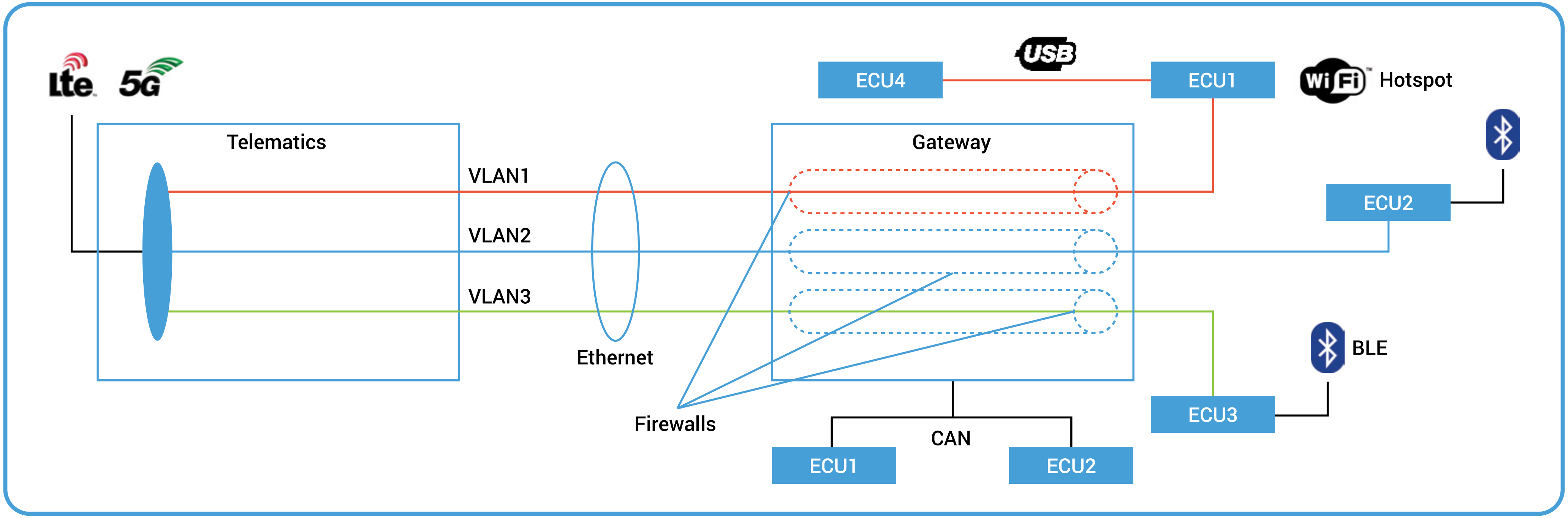



Figure 3: Connected Vehicle Architecture is followed in newer vehicles

As illustrated in figure 3, the job of the TCU is restricted to providing data paths as IP endpoints. Each of the IP endpoints creates VLANs and extends the same to the central vehicle compute unit (marked as Gateway). This gateway is a HPC (High Performance Compute) system. The Gateway system implements all the security measures and provides connectivity to other ECUs in the vehicle.

Vehicle architecture is constantly evolving with the integration of new technologies, systems, and functions. As software plays an important role in defining the vehicle functions, vehicle architecture is constantly upgraded to introduce new features, functionality and to protect it from possible security threats.

# Conclusion

With the convergence of several different technologies, including the wireless connectivity within the vehicle, software controls that define most of the functions in the vehicle, we are in the era of the 'Software Defined Vehicle'. There is a constant need to update the software throughout the life of the vehicle. In parallel with the software upgrades, there is a critical need to protect the vehicle from possible security threats and vulnerabilities. In a connected vehicle, there are over 150 ECUs that are connected and communicate with each other using multiple connectivity schemas and connect with external networks and devices. Thus, the complex vehicle architecture is vulnerable to threats from external hackers who may attempt to penetrate the vehicle network and significantly impair the functionality by gaining unauthorized control of the software. Vehicle architecture is evolving towards using high-performance compute devices and high bandwidth connectivity based on both wireless and Ethernet. Leading technology companies are collaborating with vehicle OEMs and Tier-1s to constantly improve the vehicle architecture and protect it from external vulnerabilities. In Sasken, we are working with technology leaders, vehicle OEMs, and Tier-1s to define, upgrade, and implement Cyber Safe Connected Vehicles.

# About the Author

Shantanu Das, Senior Architect in Automotive, has more than 20 years of experience working in Automotive Applications, Platform development and providing thought leadership to major automotive OEMs and Tier-1 customers all over the world. As an expert, he is instrumental in designing and developing advanced telematics control units for customers in various vehicle segments.

# About Sasken

Sasken is a specialist in Product Engineering and Digital Transformation providing concept-to market, chip-to-cognition R&D services to global leaders in Semiconductor, Automotive, Industrials, Consumer Electronics, Enterprise Devices, SatCom, Telecom, and Transportation industries. For over 32 years and with multiple patents, Sasken has transformed the businesses of 100+ Fortune 500 companies, powering more than a billion devices through its services and IP.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